

Convexity

• Polyhedron (Polyhedral sets)

Let $x \in \mathbb{R}^n$, $a \in \mathbb{R}^n$, $b \in \mathbb{R}$, A be an $m \times n$ matrix

Def hyperplane $H = \{x \mid a^T x = b\}$ eg line, plane

halfspace $H^+ = \{x \mid a^T x \geq b\}$ or $H^- = \{x \mid a^T x \leq b\}$ eg halfplane

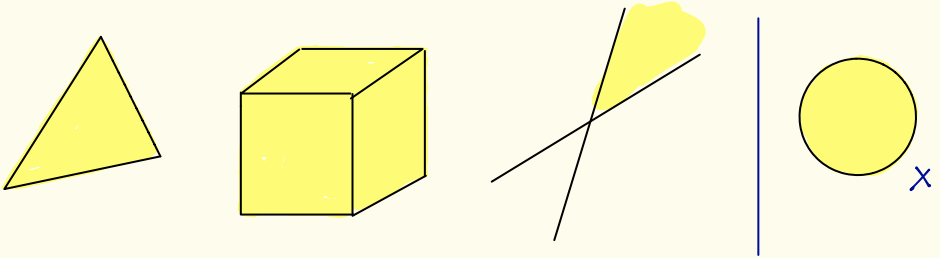
polyhedron: intersection of finite number of halfspaces

eg $S = \{x \mid Ax \geq b\}$

eg A hyperplane is a polyhedral set

eg $F = \{x \mid Ax = b, x \geq 0\}$

eg

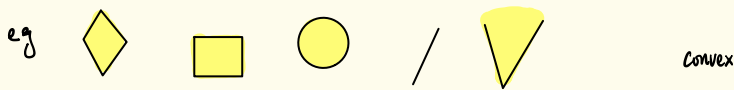


Remark: 1. A generic polyhedral set can be represented as $S = \{x \mid Ax \geq b\}$.

2. The feasible region of a linear program is polyhedral.

Convex sets

Def convex set A set S is convex if $\forall x_1, x_2 \in S$, $\underbrace{\lambda x_1 + (1-\lambda)x_2}_{\text{convex combination}} \in S \quad \forall \lambda \in [0,1]$

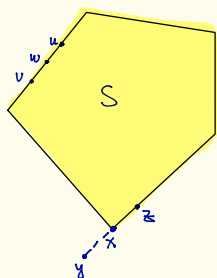


Thm Polyhedral sets are convex.

Extreme points

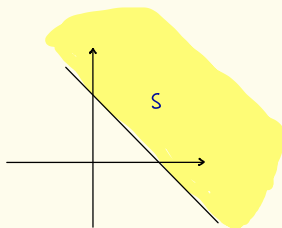
Def Let S be a convex set A vector $x \in S$ is an extreme point of S if it cannot be written as a strict convex combination of two other distinct points of S

eg

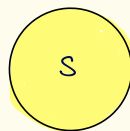


x is an extreme point

w is not



S has no extreme point



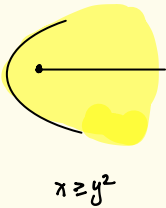
every point on the boundary is extreme.

- Directions

Def A nonzero vector $\vec{d} \in \mathbb{R}^n$ is a direction of a convex set $S \subseteq \mathbb{R}^n$ if $\vec{x} + \lambda \vec{d} \in S$ for all $\vec{x} \in S$ and $\lambda \geq 0$

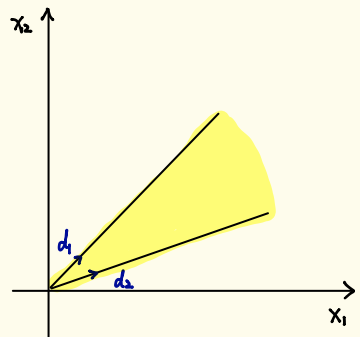
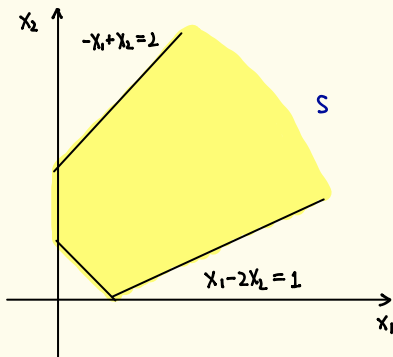
Def A direction $\vec{d}$ of a convex set S is an extreme direction if it cannot be written as a positive combination of two distinct directions of S

eg



$(1, 0)$ is a direction of $S = \{(x, y) \mid x \geq y^2\}$

eg $S = \{x \in \mathbb{R}^2 \mid -x_1 + x_2 \leq 2, x_1 + x_2 \geq 1, x_1 - 2x_2 \leq 1, x_1, x_2 \geq 0\}$



$d_1 = (1, 1)$

$d_2 = (2, 1)$